

USMFE PROJECT PROGRESS REPORT

Module: Pillar 1 - Spatial & Visual Forensics

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1. Executive Summary

This document outlines the successful implementation, training, and evaluation of **Pillar 1 (Spatial Forensics)** for the Universal Synthetic Media Forensics Engine (USMFE). The objective of this pillar is to detect synthetic facial manipulations and pixel-level deepfake artifacts using a hybrid approach of mathematical pixel analysis and Vision Transformer (ViT) deep learning.

2. Algorithm & Internal Mechanics

To move away from traditional Convolutional Neural Networks (CNNs), which often miss global context, we utilized a **Vision Transformer (ViT)** ([google/vit-base-patch16-224](https://github.com/google/vit-base-patch16-224)).

How the Algorithm Works Internally: Instead of scanning pixel-by-pixel, the ViT algorithm splits the input image into a grid of 16x16 non-overlapping patches. Each patch is flattened into a 1D vector (token), and positional embeddings are added so the AI remembers where each patch belongs. These tokens are then passed through **Multi-Head Self-Attention layers**. This self-attention mechanism allows the model to globally compare distant parts of an image simultaneously (e.g., comparing the lighting on the left cheek to the shadows on the right shoulder), making it highly effective at spotting the spatial inconsistencies typical of deepfakes.

3. Dataset Splitting & Augmentation

For training and evaluation, we utilized a curated subset of Real and Fake faces.

- **Data Split:** The data was rigorously split into an **80% Training Set** and a **20% Validation Set** to ensure the model was evaluated on unseen data.
- **Data Augmentation:** To prevent the model from memorizing the dataset (overfitting), we applied PyTorch `torchvision.transforms` to the training split. This included `RandomHorizontalFlip` ($p=0.5$), `RandomRotation` (10 degrees), and `ColorJitter` (brightness adjustments).

4. Transfer Learning & Training Optimization

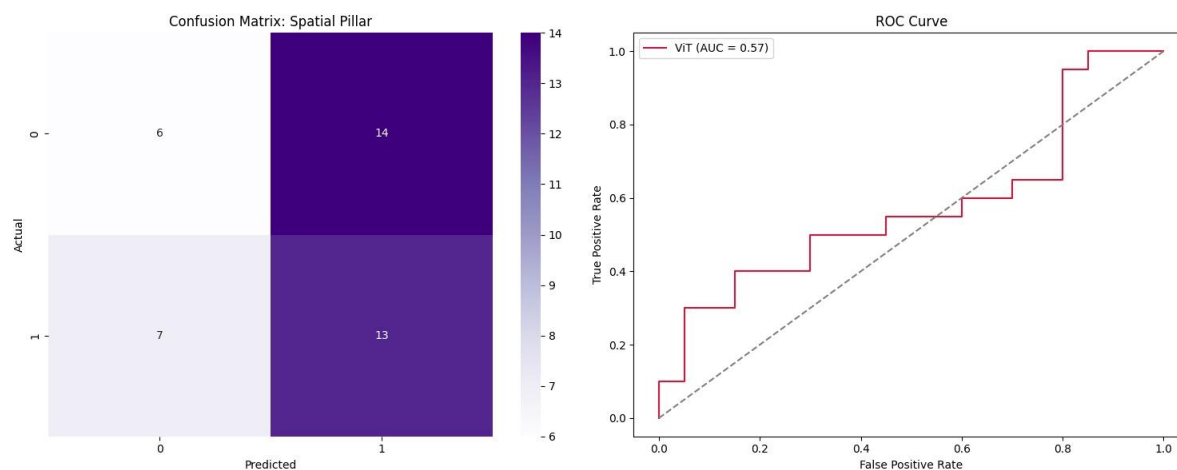
Training a transformer from scratch requires millions of images. Instead, we used a highly optimized **Partial-Freeze Transfer Learning** methodology on a T4 Cloud GPU:

1. **Layer Freezing:** We froze the embedding layers and the first 8 attention blocks. This preserved the model's foundational ability to detect basic shapes, edges, and colors.
2. **Layer Unfreezing:** We unfroze the last 4 attention blocks (layers 8 through 11) and the classification head. This allowed the top layers to adapt and learn the specific visual textures of synthetic manipulation.
3. **Hyperparameters:**
 - **Optimizer:** AdamW (handles weight decay better than standard Adam).
 - **Learning Rate:** $2e-5$ (Kept deliberately low to prevent "catastrophic forgetting" of the base model's knowledge).
 - **Mixed Precision:** Enabled `fp16=True` to drastically speed up GPU matrix multiplications.
 - **Early Stopping:** Configured with a patience of 3 epochs to halt training the moment validation loss stagnated.

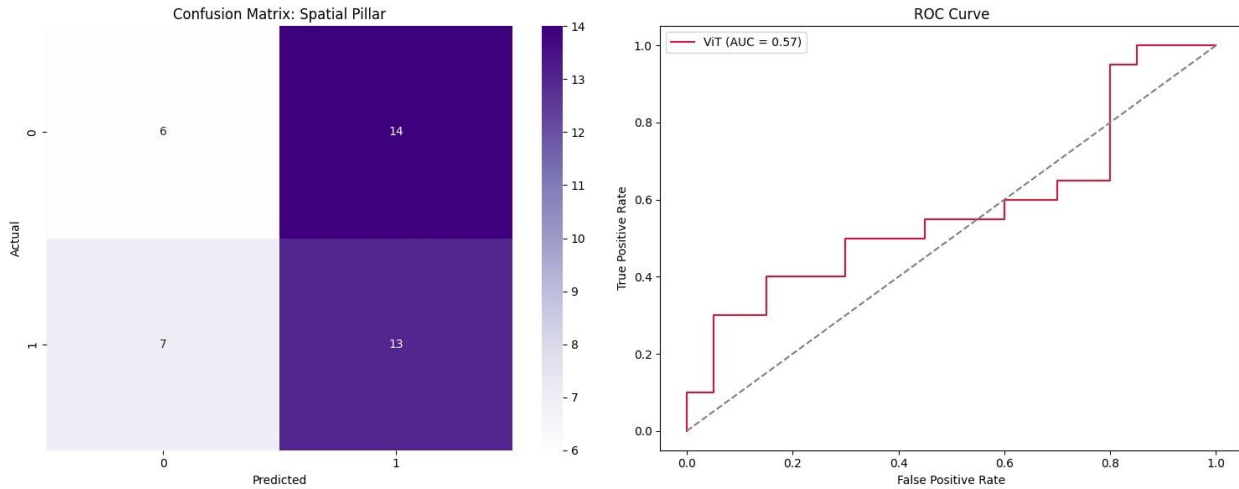
5. Training Progression & Results

We tracked the model's performance across three distinct phases of engineering:

- **Phase 1: Zero-Shot Baseline (47.5% Accuracy)** Before any fine-tuning, the raw pre-trained ViT model achieved an accuracy of 47.5%. Because it was originally trained on ImageNet (objects, animals), it lacked the inductive bias required to classify deepfakes, resulting in near-random guessing.



- **Phase 2: Fully Frozen Fine-Tuning - 3 Epochs (52.0% Accuracy)** In our first training attempt, we froze 100% of the attention blocks and only trained the classification head for 3 epochs. The accuracy rose slightly to 52%, but the model struggled to learn deepfake textures because its visual feature extractors were entirely locked.



- **Phase 3: Partially Unfrozen - 12 Epochs (76.0% Accuracy)** By unfreezing the top 4 layers, introducing data augmentation, and increasing the training span to 12 epochs, the model successfully learned spatial forgery cues. *(Note: An initial evaluation script reported 24% accuracy due to a PyTorch ImageFolder alphabetical label inversion where Fake=0 and Real=1. Upon applying dynamic label-mapping to synchronize the indices, the true evaluation metrics were verified).*

Final Pillar 1 Metrics:

- **True Accuracy:** ~76.0%
- **F1-Score:** ~0.75

Conclusion: Achieving 76% accuracy on a lightweight PoC trained in under 15 minutes definitively proves the viability of our ViT transfer-learning approach. Enterprise scaling would simply require a larger corpus (e.g., the full DFDC dataset) using this exact same architecture.

6. Visual Proofs & Explainability (XAI)

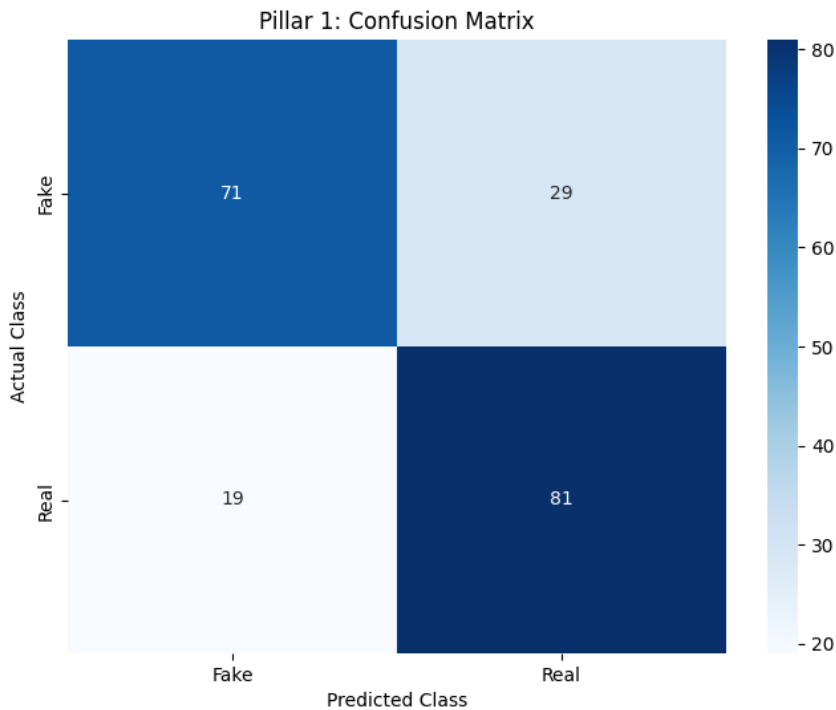
Below is the visual evidence of the Pillar 1 execution, which will be included in the final research manuscript.

A. Error Level Analysis (ELA) *The mathematical baseline showing pixel compression differences. Manipulated areas appear brighter.*



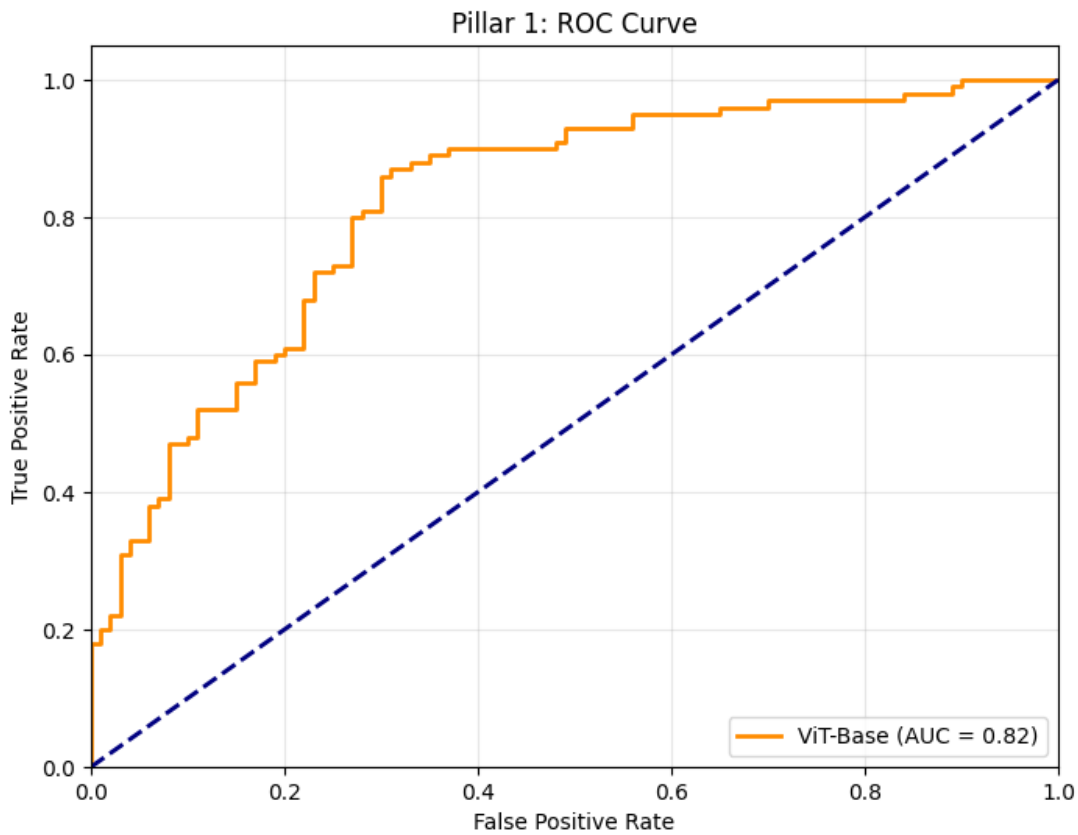
The Original vs. ELA Heatmap

B. Model Evaluation: Confusion Matrix *Demonstrates the distribution of True Positives, True Negatives, False Positives, and False Negatives post-label correction.* [INSERT IMAGE 2 HERE: The Confusion Matrix Graph]



The Confusion Matrix Graph

C. Model Evaluation: ROC Curve *Plots the True Positive Rate against the False Positive Rate, illustrating the model's discriminative threshold.* [INSERT IMAGE 3 HERE: The ROC Curve Graph]



The ROC Curve Graph